

DecodeLabs Project 1: Data Cleaning and Preparation



Dataset Overview

Every transaction in this dataset represents a customer interaction with an online business. From selecting products and making payments to applying coupons and completing orders. Like many real world datasets, it required careful preparation before analysis. To ensure accuracy and consistency, the data was reviewed for missing values, duplicate records and formatting inconsistency. After cleaning and validation, the dataset was prepared as a reliable foundation for analysis and business decision-making.

Dataset Summary

Metric	Value
Number of Rows	1200
Initial Number of Columns	14
Final Number of Columns	15
Missing values found	309
Missing Values Resolved	309
Duplicate Order IDs	0

Project Phases

Phase 1: Strategic Imputation

Issue Found	Action taken
Missing values in CouponCode	Replaced blank cells with NULL. This helped preserve records and indicated absence of coupon.

Phase 2: Integrity Audit

Check	Result
Duplicate order IDs	None found
Duplicate Records	None found

Phase 3: Speak One Language

Transformation	Purpose
Split DateTime into Date and Time	Improved analysis
Trimmed text	Removed leading or trailing spaces
Cleaned text	Removed hidden characters
Capitalized each word	Standardized text
Currency formatting	Standardized monetary values

Conclusion

Following data cleaning and validation, the dataset was transformed into a consistent and analysis-ready format. The final dataset contains 1200 rows and 15 columns, with missing values addressed appropriately, no duplicate Order IDs identified, and all key fields standardized for accuracy and consistency. This ensures the dataset is reliable for further analysis and decision-making.

Dataset before cleaning

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	285.
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	30.
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	275.
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.
ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.
ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.
ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.
ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2		Email	268.
ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.
ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.
ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.
ORD200012	2024-10-15	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	3
ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.
ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.
ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3		Google	473.
ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.
ORD200017	2024-03-02	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	TRK11972116	3		Referral	42.
ORD200018	2025-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook	431.
ORD200019	2023-01-23	C14552	Monitor	5	224	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	11
ORD200020	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.

Cleaning Queries

Queries [1]
Table1

Table.TransformColumns(#"Cleaned Text",{{"OrderID", Text.Proper, type text}, {"CustomerID", Text.Proper, type

OrderID	Date	Time	CustomerID	Product	Quantity
Ord200000	04/01/2023	12:00:00 AM	C72649	Monitor	
Ord200001	23/08/2024	12:00:00 AM	C75739	Phone	
Ord200002	27/02/2024	12:00:00 AM	C81728	Tablet	
Ord200003	15/10/2023	12:00:00 AM	C33540	Chair	
Ord200004	08/05/2025	12:00:00 AM	C81840	Printer	
Ord200005	23/10/2023	12:00:00 AM	C37249	Phone	
Ord200006	17/06/2025	12:00:00 AM	C83492	Laptop	
Ord200007	12/05/2023	12:00:00 AM	C41460	Monitor	
Ord200008	02/04/2025	12:00:00 AM	C26817	Phone	
Ord200009	21/11/2023	12:00:00 AM	C31946	Desk	
Ord200010	29/12/2023	12:00:00 AM	C43443	Tablet	
Ord200011	17/02/2024	12:00:00 AM	C93861	Monitor	
Ord200012	15/10/2024	12:00:00 AM	C38785	Monitor	
Ord200013	30/08/2023	12:00:00 AM	C88348	Laptop	
Ord200014	27/03/2023	12:00:00 AM	C98474	Tablet	
Ord200015	17/07/2023	12:00:00 AM	C39416	Printer	
Ord200016	24/01/2023	12:00:00 AM	C28976	Printer	
Ord200017	02/03/2024	12:00:00 AM	C60672	Tablet	
Ord200018	26/03/2025	12:00:00 AM	C84329	Desk	
Ord200019	23/01/2023	12:00:00 AM	C14552	Monitor	
Ord200020	08/05/2023	12:00:00 AM	C54829	Chair	
Ord200021	17/11/2024	12:00:00 AM	C99023	Monitor	
Ord200022	29/01/2023	12:00:00 AM	C34416	Monitor	

15 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name
Table1

All Properties

APPLIED STEPS

Source

Changed Type

Split Column by Delimiter

Changed Type1

Renamed Columns

Removed Duplicates

Changed Type2

Replaced Value

Trimmed Text

Cleaned Text

Capitalized Each Word

Dataset after cleaning

A1	OrderID													
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	OrderID	Date	Time	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource
2	Ord200000	2023-01-04	12:00:00 AM	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	Trk37947903		7 Save10	Instagram
3	Ord200001	2024-08-23	12:00:00 AM	C75739	Phone	2	151.35	823 Main St	Online	Shipped	Trk91186779		3 Save10	Referral
4	Ord200002	2024-02-27	12:00:00 AM	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	Trk42903982		8 Freeship	Email
5	Ord200003	2023-10-15	12:00:00 AM	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	Trk62788070		5 Save10	Facebook
6	Ord200004	2025-05-08	12:00:00 AM	C81840	Printer	4	626.01	668 Main St	Online	Delivered	Trk29241424		8 Save10	Email
7	Ord200005	2023-10-23	12:00:00 AM	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	Trk72976927		4 Save10	Instagram
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10	Ord200008	2025-04-02	12:00:00 AM	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	Trk61042692		2 Null	Email
11	Ord200009	2023-11-21	12:00:00 AM	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	Trk33478363		6 Save10	Google
12	Ord200010	2023-12-29	12:00:00 AM	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	Trk98859248		9 Winter15	Instagram
13	Ord200011	2024-02-17	12:00:00 AM	C93861	Monitor	3	49.14	831 Main St	Online	Returned	Trk48234646		7 Save10	Email
14	Ord200012	2024-10-15	12:00:00 AM	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	Trk20419991		6 Freeship	Referral
15	Ord200013	2023-08-30	12:00:00 AM	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	Trk81913874		4 Freeship	Facebook
16	Ord200014	2023-03-27	12:00:00 AM	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	Trk79186539		3 Save10	Instagram
17	Ord200015	2023-07-17	12:00:00 AM	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	Trk54930938		3 Null	Google
18	Ord200016	2023-01-24	12:00:00 AM	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	Trk13194259		3 Freeship	Email
19	Ord200017	2024-03-02	12:00:00 AM	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	Trk11972116		3 Null	Referral
20	Ord200018	2025-03-26	12:00:00 AM	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	Trk65191045		3 Save10	Facebook
21	Ord200019	2023-01-23	12:00:00 AM	C14552	Monitor	5	224	273 Main St	Cash	Returned	Trk59101365		9 Save10	Google
22	Ord200020	2023-05-08	12:00:00 AM	C54829	Chair	5	62.51	587 Main St	Cash	Pending	Trk16718651		9 Winter15	Instaeram
	<	>	Raw Data	Cleaned Data	Documentation	+								